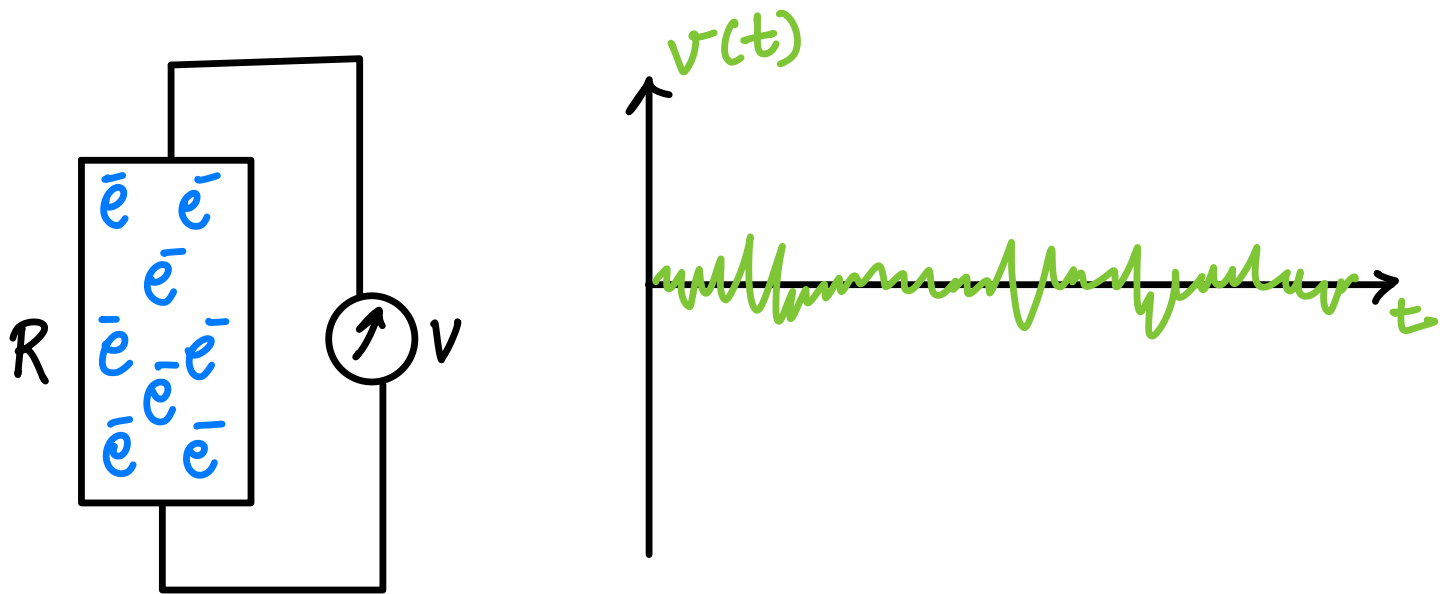




Noise



Microstate: Distribution of electrons in the resistor

Sample space: Set of all microstates (Ω)

Random variable

A function that assigns a real number to each microstate.

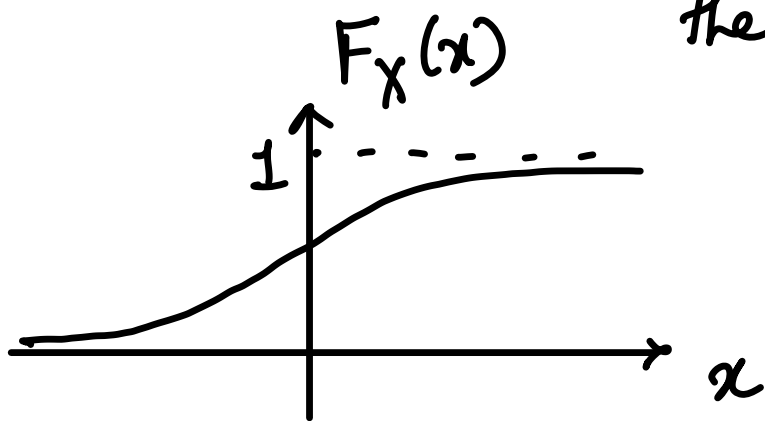
$$X: \Omega \rightarrow \mathbb{R}$$

loosely speaking $V(t) \in \mathbb{R}$ is called the RV.

Cumulative Distribution Function (CDF).

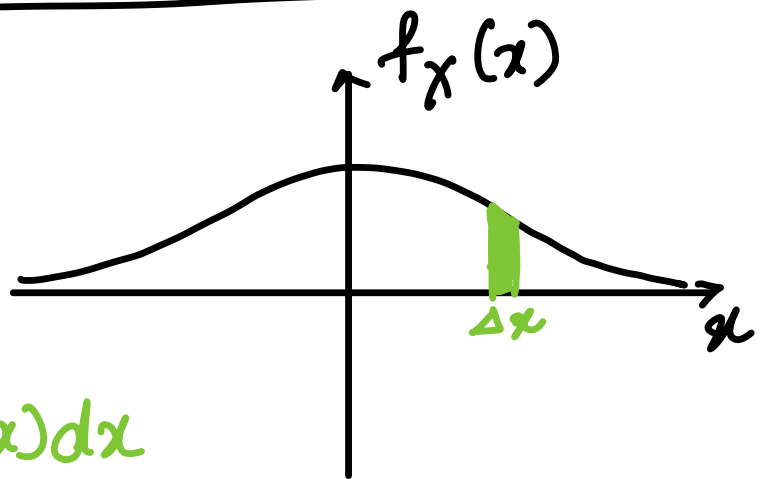
$$F_X(x) \triangleq P(X \text{ is less than } x)$$

Say $v = 0.1 \rightarrow$ Do a large number of experiments & see what is the prob. that $X < x$.



Probability Density Function (PDF)

$$f_X(x) = \frac{dF_X(x)}{dx}$$

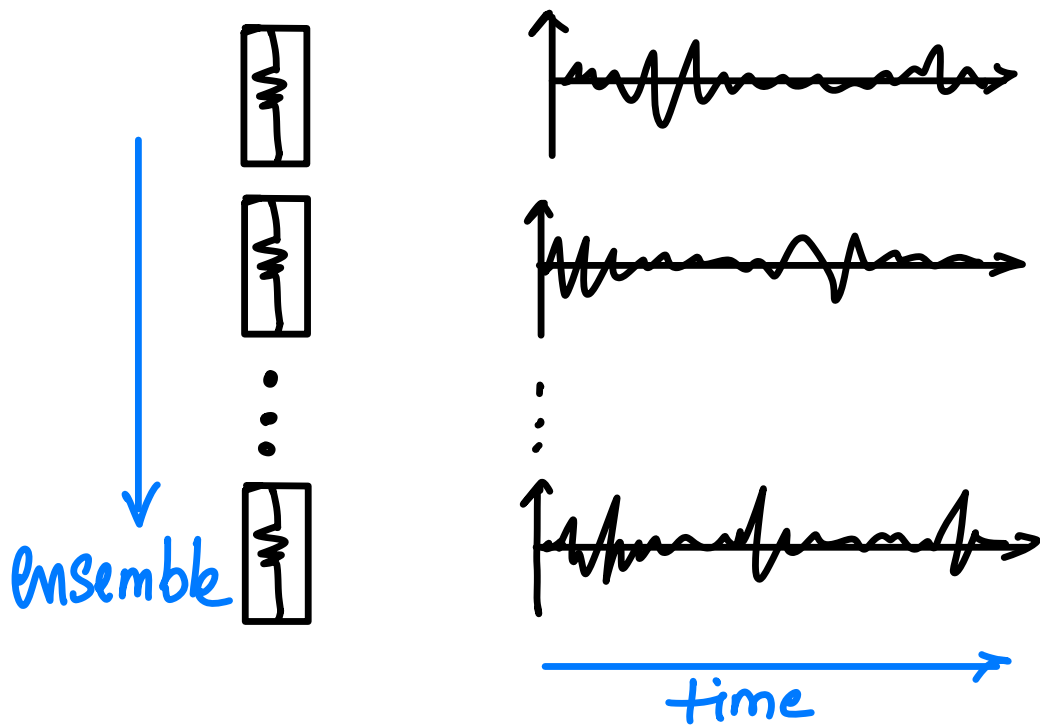


$$P(a < x < b) = \int_a^b f_X(x) dx$$

How concentrated is the Probability in the nbd of x .

Random Process

$X(t)$: A collection of RVs indexed by time.



Statistics of Random Variables

$$E[X] = \int_{-\infty}^{\infty} x f_X(x) dx \rightarrow \text{weighted average.}$$

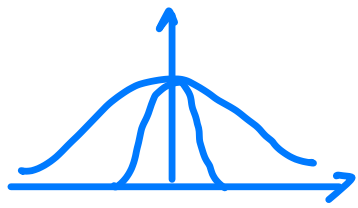
$$E[g(X)] \triangleq \int_{-\infty}^{\infty} g(x) f_X(x) dx$$

Linearity: $E[aX + bY] = aE[X] + bE[Y]$

Moments

mean: $\mu_x = E[X] = \int_{-\infty}^{\infty} x f_X(x) dx$

Variance: $\sigma_x^2 = E[(X - \mu_x)^2]$



$$= \int_{-\infty}^{\infty} (x - \mu_x)^2 f_X(x) dx$$

$$= E[X^2] - \mu_x^2$$

Central
Moment of
order n

$$E[(X - \mu_x)^n]$$

3 $\rightarrow$ Skew

4 $\rightarrow$ Kurtosis

Covariance: $\text{Cov}(X, Y) = E[(X - \mu_x)(Y - \mu_y)]$
 $= E[XY] - \mu_x \mu_y.$

"How well do X, Y track each other".

$\text{Cov}(X, Y) = 0 \Rightarrow X, Y$ are uncorrelated.

$$\text{Cov}(X, X) = \sigma_X^2$$

Correlation coefficient $\rho_{xy} = \frac{\text{Cov}(X, Y)}{\sigma_X \sigma_Y}$

$$\rho_{xy} \in [-1, 1]$$

Central Limit Theorem

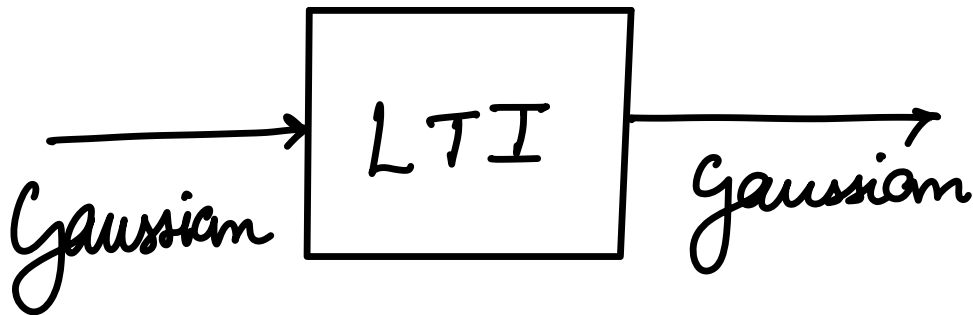
Given n independent distributions $X_1, X_2, \dots, X_n$; their sum $\sum_{m=1}^n X_m$

converges to a Gaussian distribution as $n \rightarrow \infty$.

$\Rightarrow$ Thermal noise is Gaussian in distribution

$$f_X(x) = \frac{1}{\sqrt{2\pi} \sigma_X} \exp\left(-\frac{(x - \mu_X)^2}{2 \sigma_X^2}\right) \sim \mathcal{N}(\mu_X, \sigma_X^2)$$

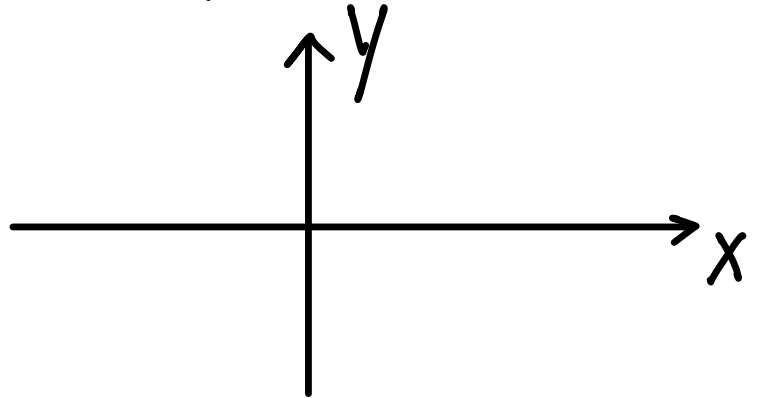
- ▷ Closed under linearity & linear systems.



- ▷ Uncorrelated $\Leftrightarrow$ Independent

$$\rho_{xy} = 0$$

$$f_{xy}(x, y) = f_x(x)f_y(y)$$



- ▷ Typically $\mu_x = 0$ for noise.

- ▷ Fully determined by mean, variance & time statistics or freq. statistics.

Statistics of Random Processes

Auto-correlation

$$R_{XX}(t_1, t_2) = E[X(t_1)X(t_2)]$$

"How similar is X to itself at t_1 & t_2 ".

Stationary (Wide-sense)

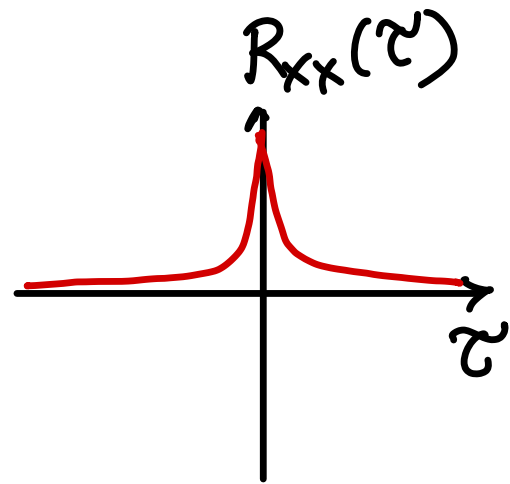
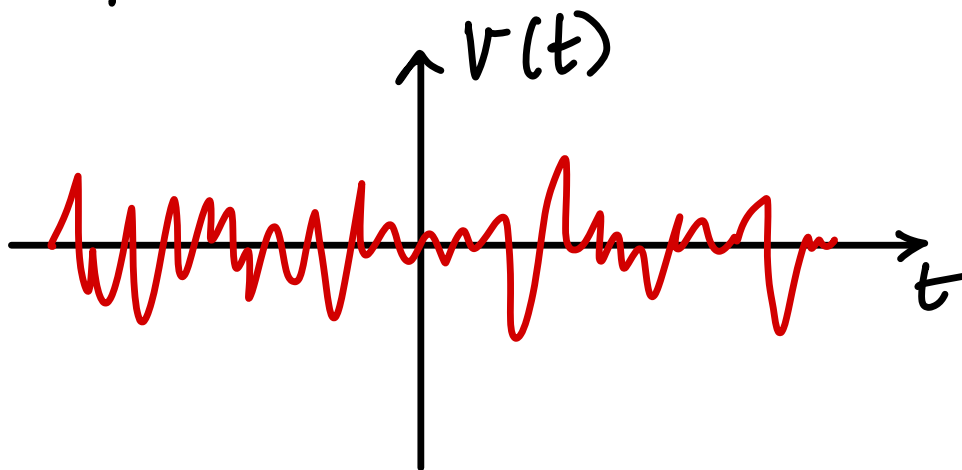
▷ Statistical properties do not change with time.

$$▷ \mu_X(t) = \mu_X$$

$$\begin{aligned}▷ R_{XX}(\tau) &= E[X(t)X(t+\tau)] \\ &= \int_{-\infty}^{\infty} \int_{-\infty}^{\infty} x_1 x_2 f_{X(t)X(t+\tau)}(x_1, x_2) dx_1 dx_2.\end{aligned}$$

$$▷ R_{XX}(0) = E[X^2] = \sigma_X^2$$

$$\triangleright |R_{xx}(\tau)| \leq |R_{xx}(0)|$$



Ergodicity Time average = Ensemble average.

$$\mu_x = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) dt$$

$$\sigma_x^2 = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x^2(t) dt$$

$$R_{xx}(\tau) = \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} x(t) x(t+\tau) dt$$

$v(t) \rightarrow$ instantaneous voltage (rms).

$p(t) = \frac{v^2(t)}{R} \rightarrow$ instantaneous power

$$\begin{aligned} P_{av}(t) &= \lim_{T \rightarrow \infty} \frac{1}{T} \int_{-\frac{T}{2}}^{\frac{T}{2}} \frac{v^2(t)}{R} dt \\ &= \frac{\overline{v^2}}{R} \quad \text{time average.} \\ &= \frac{\overline{v^2}}{R} \end{aligned}$$

$$\begin{aligned} \text{Power} &= E[(X+Y)^2] = E[X^2] + E[Y^2] \\ &\quad + 2E[\cancel{XY}] \rightarrow 0 \\ &= 2E[X^2] \quad \text{if } \sigma_x = \sigma_y \end{aligned}$$

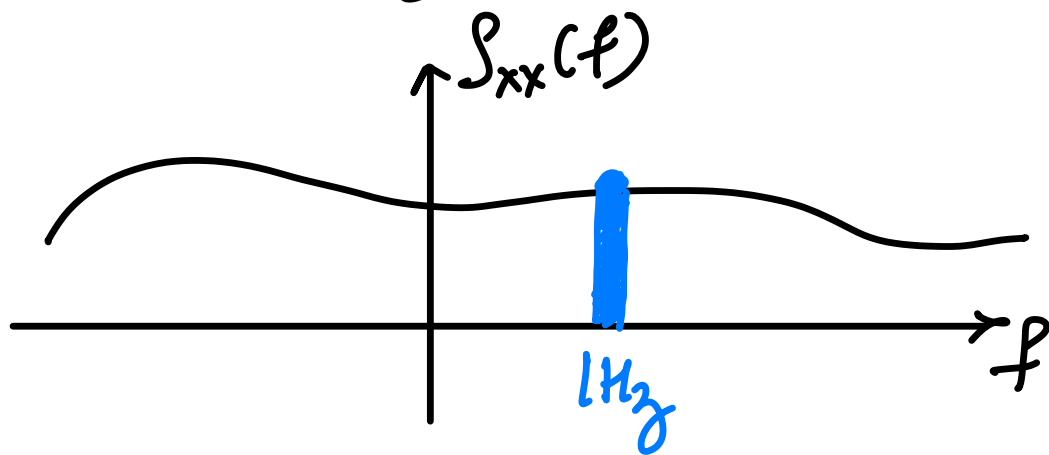
$$\begin{aligned} \text{If correlated} &\Rightarrow E[(2X)^2] \\ &= 4E[X^2] \end{aligned}$$

$$\rho_{xy} = 1 \Rightarrow 4x$$

$$\rho_{xy} = -1 \Rightarrow 0$$

Power Spectral Density (PSD)

$S_{xx}(f)$: How much power is in a 1Hz bandwidth centered at f .



$$S_{xx}(f) = \lim_{\Delta f \rightarrow 0} \frac{P_x(\Delta f)}{\Delta f}$$

Wiener Khinchin Thm

For a WSS process, the autocorrelation and the PSD form a FT pair.

$$S_{xx}(f) = \int_{-\infty}^{\infty} R_{xx}(\tau) e^{-j2\pi f\tau} d\tau$$

$$R_{xx}(\tau) = \int_{-\infty}^{\infty} S_{xx}(f) e^{+j2\pi f\tau} df$$

$$\int_{-\infty}^{\infty} S_{xx}(\omega) d\omega = R_{xx}(0) = E[x^2] = \sigma_x^2$$

White Noise: $S_{xx}(\omega) = N_0$

$$\Rightarrow R_{xx}(\tau) = N_0 \delta(\tau)$$

Noise in Linear Systems

$$S_{yy}(\omega) = |H(\omega)|^2 S_{xx}(\omega)$$



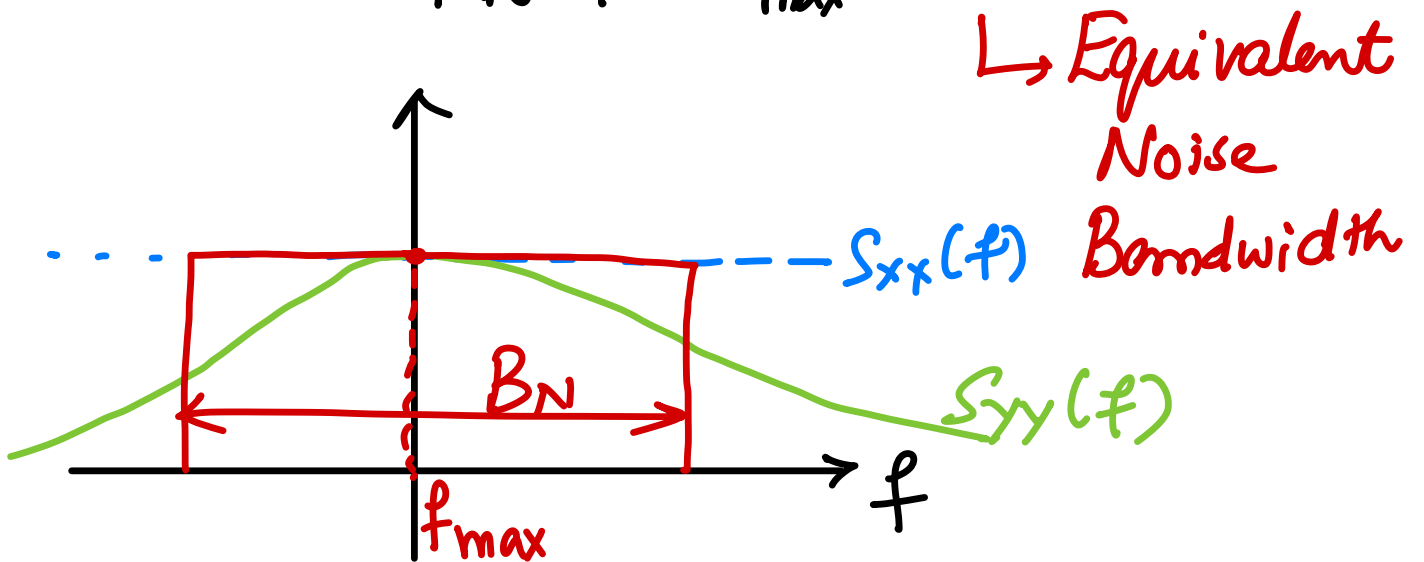
$$E[Y^2] = \int_{-\infty}^{\infty} S_{yy}(\omega) d\omega$$

$$= \int_{-\infty}^{\infty} |H(\omega)|^2 S_{xx}(\omega) d\omega$$

If input noise is white $S_{xx}(\omega) = N_0$

$$\Rightarrow E[y^2] = N_0 \int_{-\infty}^{\infty} |H(f)|^2 df$$

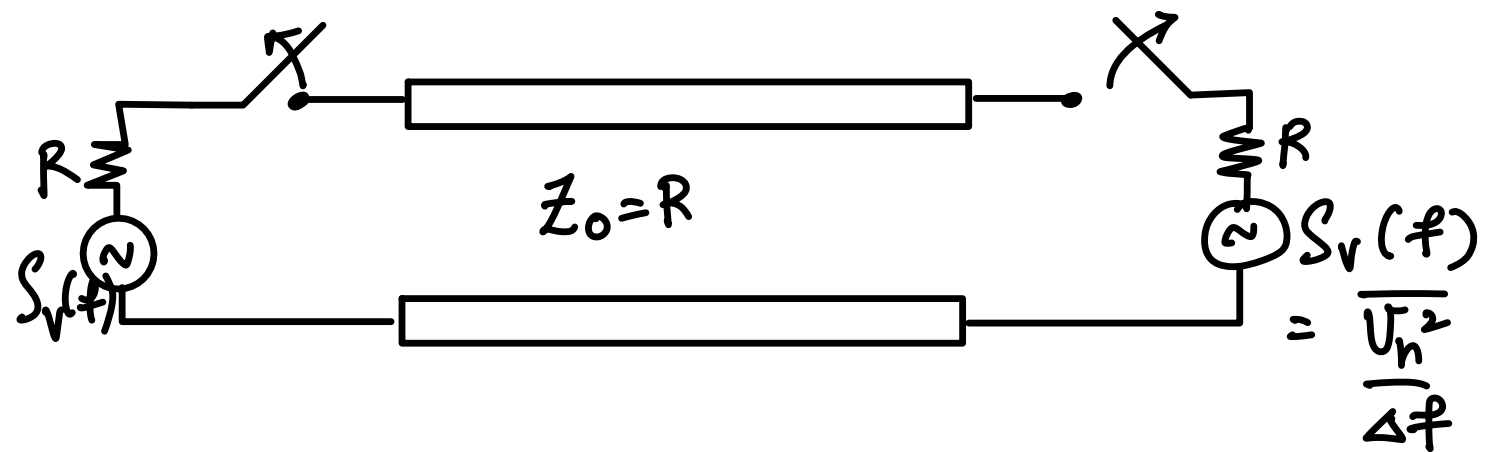
$$= N_0 |H(f)_{\max}|^2 B_N$$



$$B_N = \frac{\int_{-\infty}^{\infty} |H(f)|^2 df}{|H(f_{\max})|^2}$$

$$B_N \approx \frac{\pi}{2} f_{-3dB} \text{ for a single pole LPF.}$$

Thermal Noise (Nyquist)



► Power delivered to the Transmission line from each source is $\frac{\overline{V_n^2}}{4R}$

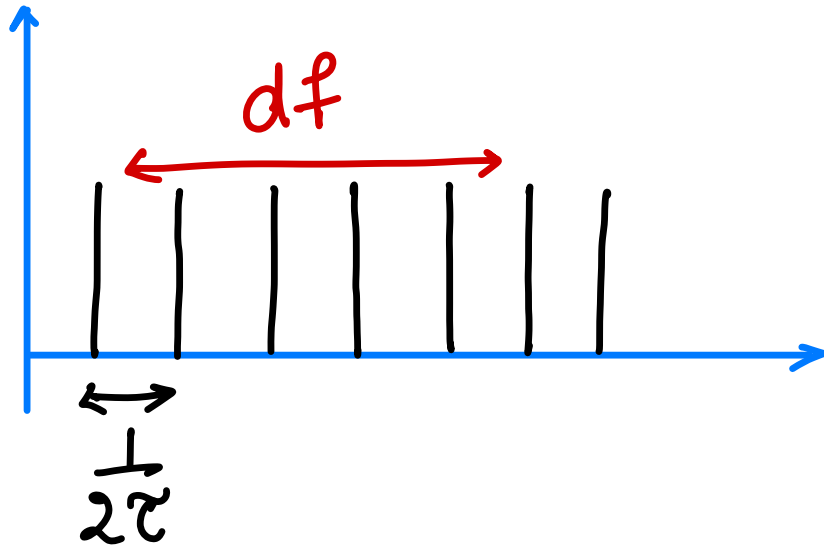
► Total energy in the T-line is

$$2 \cdot \tau \cdot \frac{\overline{V_n^2}}{4R}$$

$$dE = 2\tau dP = 2\tau \frac{\overline{V_n^2}}{4R} = \frac{2\tau}{4R} S_v(f) df$$

$$\Rightarrow dE = \frac{2\tau}{4R} S_v(f) df \quad \text{--- (1)}$$

▷ Mode spacing is $\frac{1}{2\tau}$ $\frac{\lambda}{2}, 1, \frac{3\lambda}{2}, \dots$



⇒ No. of modes in $d\nu = 2\tau d\nu$

▷ Equipartition Thm:

Each mode carries an energy of

$$\frac{h\nu}{e^{\frac{h\nu}{kT}} - 1} \xrightarrow[h\nu \ll kT]{h\nu \ll kT} kT.$$

⇒ $\nu < 10TH_z$

$$dE = (2\tau d\nu)(kT) \quad \text{--- (2)}$$

$$\cancel{2\tau d\nu} kT = \frac{\cancel{2\tau}}{4R} S_\nu(\nu) \cancel{d\nu}$$

$$S_v(f) = 4kTR$$

$$S_v(f) = 4kTR \left(\frac{\frac{hf}{kT}}{e^{\frac{hf}{kT}} - 1} \right) \approx \frac{4kTR}{1 + \frac{1}{2} \frac{hf}{kT}}$$

Noise power delivered to a matched load

$$P_{\text{del}} = \frac{\overline{V_n^2}}{4R} = \frac{4kTRB}{4R} = kTB$$

$$P_{\text{del}} = kTB$$